

AAI-510 · TEAM 7 · UNIVERSITY OF SAN DIEGO

LexPath AI

Intelligent Legal Intake & Case Routing

June 2026



Christina Sadiq

Project Manager

Nicole Rowley

AI Engineer

Marie Angelique Membrido

Data Engineer

THE BUSINESS PROBLEM

The Intake Bottleneck



Attorneys manually screen every prospect call



2+ hrs/week per attorney lost to unqualified leads



No conflict-of-interest check before engagement



Routing errors reach the wrong practice team

AT RISK ANNUALLY

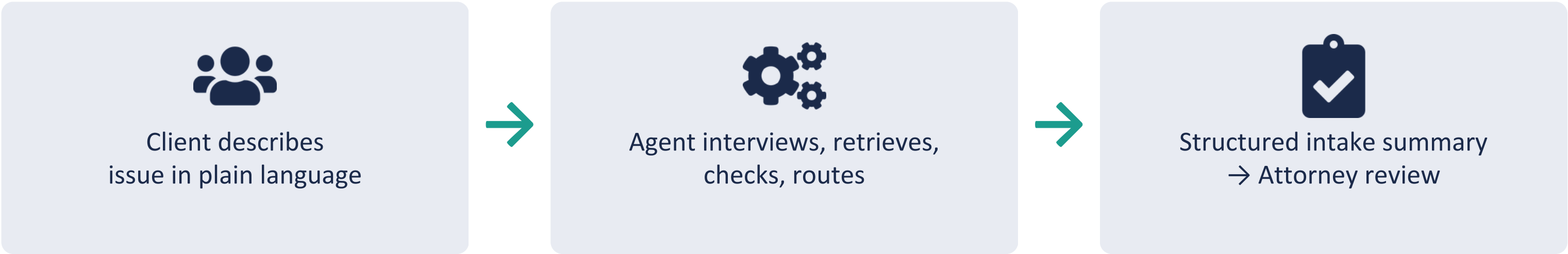
\$1.56M

in recoverable attorney capacity

50 attorneys · \$300/hr · 2 hrs/week lost

SOLUTION AT A GLANCE

How LexPath AI Works



ReAct pattern: Reason → Act → Observe → repeat

 **Semantic Retrieval**

 **Conflict Check**

 **Case Routing**

Every output requires senior attorney approval

No matter proceeds without attorney approval

EVERY OUTPUT

```
"status": "READY_FOR_REVIEW"
```

```
"hitl_flag": true
```

Liability stays with the attorney. The agent screens; it never decides.

1

Conflict review

2

Routing validation

3

Senior partner sign-off

The Business Case



\$1.56M/yr max recoverable capacity · *50 attorneys, \$300/hr, 2 hrs/week, 40 intakes/week*

- **Claude 4.6:** 48% routing accuracy → about \$749K/yr in effective business value
- **GPT-4.1:** 88% routing accuracy → about \$1.37M/yr in effective business value
- **LLM cost difference:** \$16 vs \$10 per year — a rounding error either way
- **Net advantage:** GPT-4.1 delivers roughly \$624K more business value per year
- **ROI multiple:** ~46,000x for Claude, ~144,000x for GPT-4.1 — both pay for themselves in week one

Bottom line: the gap is driven almost entirely by routing accuracy, not token cost.

LEDGAR Dataset

~60,000

SEC EDGAR contractual provisions

100

labeled legal categories (LexGLUE benchmark)

DATABRICKS PIPELINE



Delta Table



Embeddings



Vector Search Index

Eval set: 50-row held-out test split · fixed seed 42



Three Tools, One Pipeline

1



Semantic Retrieval

Queries LEDGAR Vector Search → top-5 matching provisions

2



Conflict Check

Searches internal client DB →
CLEARED or CONFLICT_FLAG

3



Case Routing

Maps category to practice area:
Corporate, Litigation, Employment, IP,
Real Estate



Deployment Recommendation

1

Supervised Pilot

Months 1–3

- 1 office
- Paralegal reviews every output



2

Expanded

Months 4–9

- All offices
- Review only flagged cases



3

Full Production

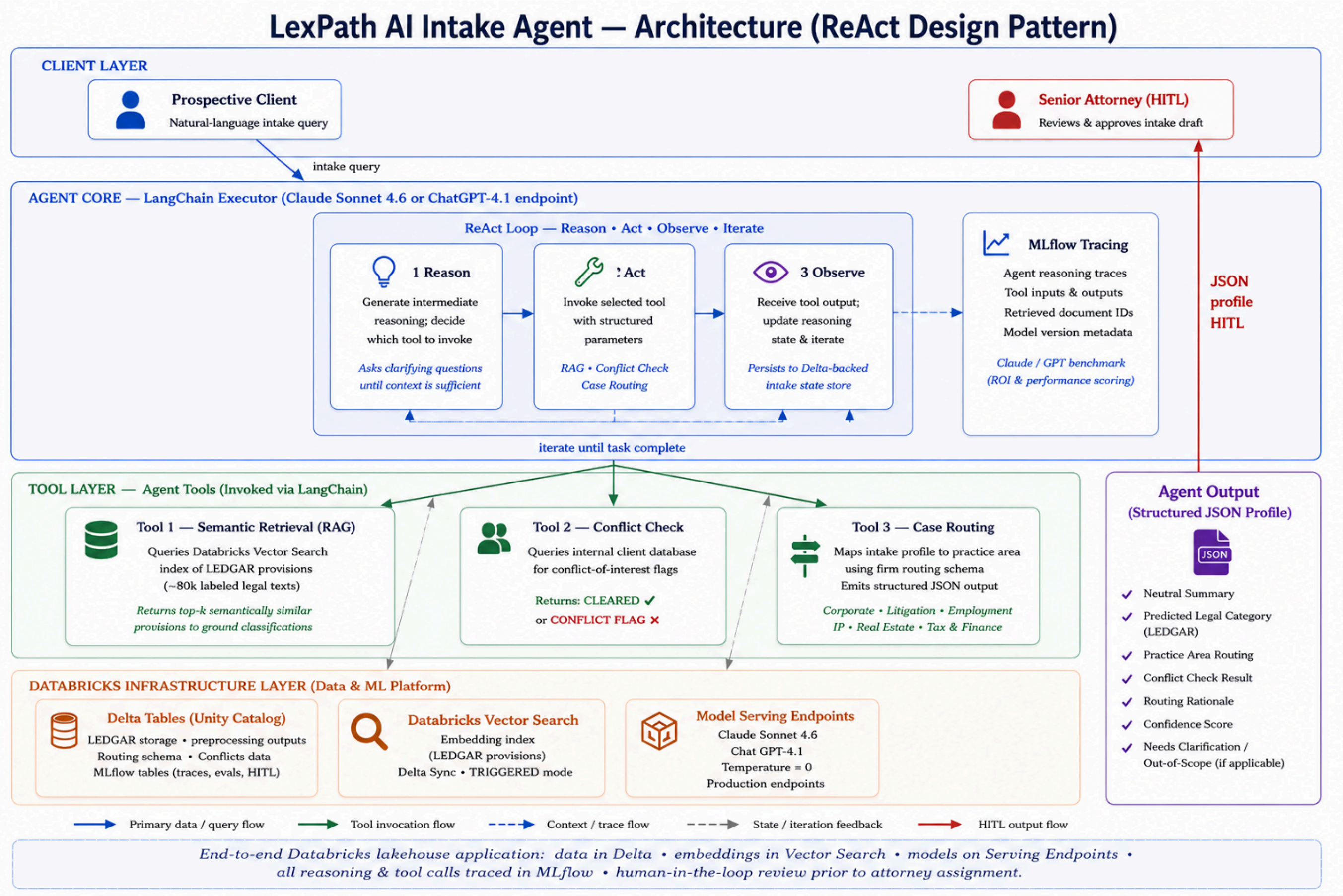
Month 12+

- Integrates with case mgmt system
- Corrections update routing

Full deployment is out of scope for this project; the architecture supports this path.



TECHNICAL ARCHITECTURE



Everything runs from a shared agent_lib.py so build, run, and evaluation notebooks stay identical.

Five MLflow Benchmark Traces

Tracing with MLflow provided end-to-end visibility into agent decision-making and workflow execution.

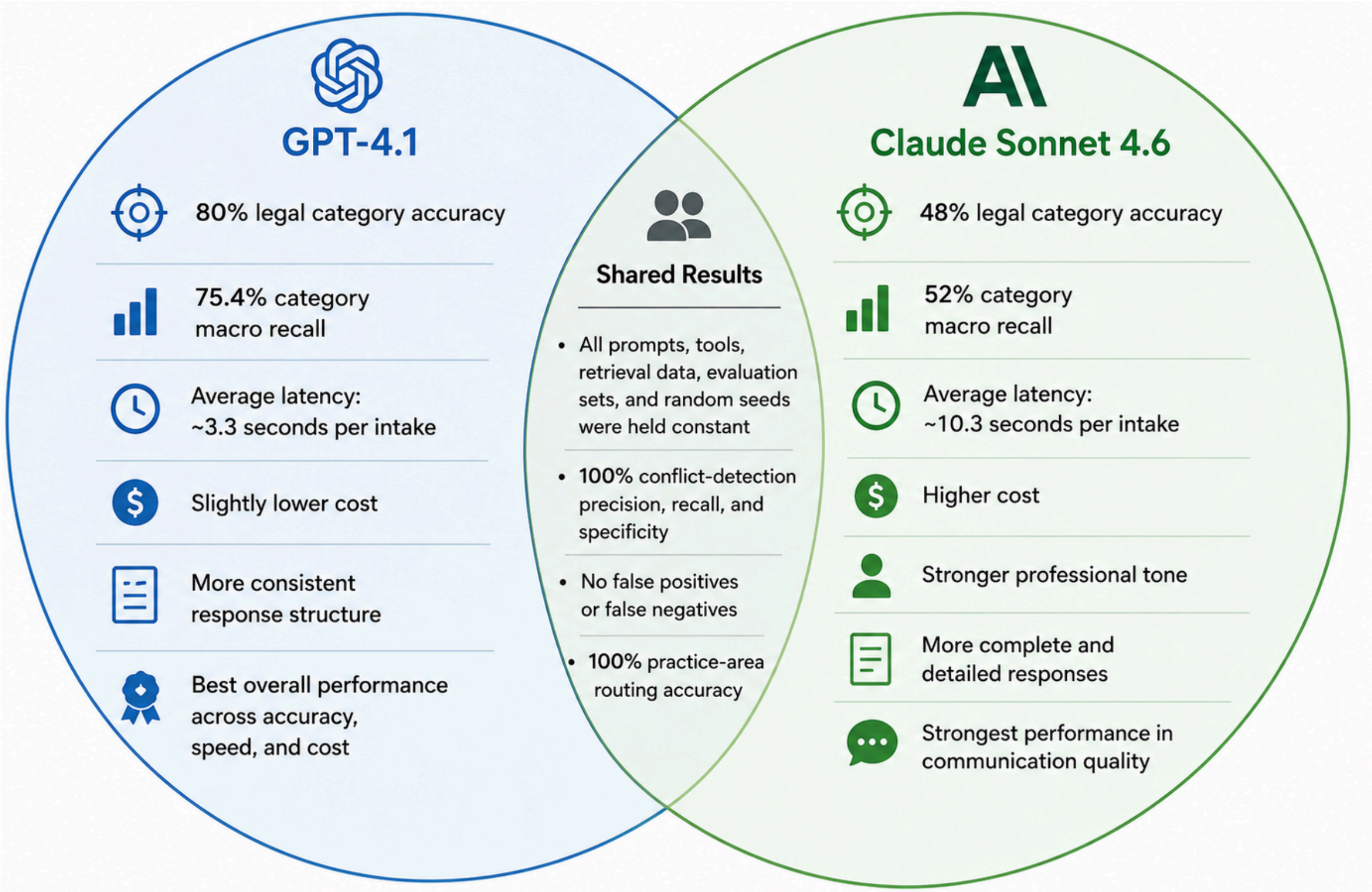
Five benchmark scenarios evaluated different workflow behaviors:

1	Standard arbitration matter	 Litigation
2	Conflict-detection case: Atlas Manufacturing	 CONFLICT_FLAG
3	Employment dispute	 Employment
4	Ambiguous intake requiring clarification	 Clarifying questions
5	Out-of-scope legal advice request	 Graceful rejection

+ Traces revealed two primary execution patterns:

- Single-step responses for rejections and clarification requests
- Multi-step workflows involving multiple tool calls for full legal intakes

Claude Sonnet 4.6 vs GPT-4.1



Evaluation Beyond Accuracy

- Effective agent evaluation requires multiple perspectives beyond accuracy metrics.
- Evaluation included legal compliance, completeness, professional tone, response structure, tool usage, and data grounding.
- Practice-area routing outperformed exact category accuracy, highlighting the value of business-focused metrics.
- This framework assessed both answer quality and adherence to legal and operational requirements.
- Although both models achieved perfect legal compliance, qualitative review revealed meaningful differences in response style and organization.
- The results highlight the importance of combining quantitative metrics with qualitative evaluation in professional AI deployments.

Human Feedback and Alignment

- Evaluation progressed from manual review to automated LLM judges
- Human oversight is essential for setting objectives, validating outcomes, and correcting evaluation drift.
- Agentic systems should augment, not replace, professional judgment.
- Agents are best suited for repetitive administrative tasks, allowing experts to focus on higher-value decisions.
- In legal services, trust depends on transparency, accountability, and human oversight rather than full autonomy.

Key Insights and Future Work

- 💡 RAG effectively grounded decisions in retrieved legal provisions.
- 💡 Exact classification across 100 categories remained challenging.
- 💡 Practice-area routing outperformed exact category accuracy, highlighting the value of business-focused metrics.
- 💡 MLflow tracing, automated judges, and human review provided deeper insights than aggregate metrics alone.
- 💡 Future work: larger conflict datasets, expanded evaluations, continuous monitoring, and case-management integration.
- 💡 Effective agentic AI combines retrieval, structured workflows, rigorous evaluation, and human oversight to augment professional expertise.

LexPath AI

Next up: demonstration

Watch a demo of the data pipeline → agent workflow → MLflow traces → LLM as a Judge and human review process in practice.

```
github.com/nicole-rowley-msaai/MS_AAI_510_Agentic_AI_Sys_Final_Project
```

Christina Sadiq · Nicole Rowley · Marie Angelique Membrido | University of San Diego | AAI-510 | June 2026

